

Reinforcement Learning-Aided UAV Relay with Buffer for Outage Probability Minimization

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Abstract—The deployment of an Unmanned Aerial Vehicles (UAVs) as relays has prompted substantial interest in 6G mobile wireless networks. Nevertheless, due to high UAV's altitude path loss and multipath fading, the direct transmission of data by non-buffering UAV could lead to an elevated Outage Probability (OP). Thus, signal and packet loss are the key challenges that undermine consistent connectivity and remain to be addressed. To address these limitations, this paper explores the integration of a buffer mechanism-aided UAV relaying system to enhance the reliability by managing data flow. Furthermore, we propose Reinforcement Learning (RL) approach to intelligently optimize the UAV relay position relative to the Base Station (BS) and Ground User (GU). Specifically, we consider a framework where UAV relay aided by both buffer mechanism and Q-learning to minimize the overall OP. Simulation results reveals that the proposed approach consistently outperforms the baseline schemes, demonstrating enhanced reliability and convergence.

Index Terms—6G, UAV Communications, Reinforcement Learning, Q-learning, Outage Probability

I. INTRODUCTION

The integration of Unmanned Aerial Vehicles (UAVs) have rapidly evolved into adaptable platforms as a promising solution, extending the network coverage, enhancing signal quality, and improving overall reliability in a variety of applications [1]. Owing to their mobility and adaptability, UAVs can be exploited as aerial relays between ground infrastructure and user equipment. Therefore, it is essential to address challenges such as signal degradation and packet loss owing to deteriorating channel conditions or errors and interference. Aerial relays possess the distinctive capability of adjusting their height while enhancing system performance in contrast to traditional terrestrial infrastructures (e.g., ground relays) [2]. Thorough study has been undertaken on system design, suitable channel configurations, and optimization challenges for UAV-assisted networks. For instance, UAV exploits as a mobile relay between a source and destination for maximizing system throughput utilizing the amplify-and-forward protocol [3], where the studies in [4], [5] analyze the effect of interference on UAV communications and co-existence of a ground user and UAV for establishing control and data links by considering the Line-of-Sight (LoS) probability and different channel fadings for LoS and Non-NLoS links, which are

influenced by the elevation angle of the communication link. An investigation of UAV relay to guarantee reliability, considering Nakagami-m [6], a combination of fading channels to mitigate the potential of encountering Outage Probabilities (OPs) was examined [7]. However, [3]–[5], [7] companies carried out their endeavors by considering direct signals. Nevertheless, buffering strategies were not extensively presented in literature. To further enhance the performance of mobile wireless networks, a buffer mechanism was introduced to maximize the Energy Efficiency (EE) in [8], and optimize the data throughput in [9], [10]. However, [8]–[10] utilizing the adaptability of aerial relay and buffer mechanisms without directly referencing OP. Another critical challenge is finding the optimal UAV position that can minimize the path loss that leads to minimizing the OP. However, the majority of the studies conducted based on the classical approaches to find the optimal position of UAVs either exploit a one-dimensional search [11], two-dimensional search leveraging the Golden Search method [12], or a trade-off approach to 3D position [7]. Reinforcement Learning (RL) based on the Q-learning approach can be exploited to optimize the UAV's position efficiently [13]. Extensive research with a focus on UAV's position optimization and resource allocation have explored RL based UAVs [14]–[17]. The study in [14] investigated a distributed RL based framework for EE efficiency, subsequent works, such as [15], investigated a Q-learning-based UAV in ad hoc networks to reduce error rates and enhance communication performance against smart jammers, while [16] explored Deep RL to resist smart jamming. Additionally, RL was investigated for optimizing UAV's trajectory in [17]. Despite the aforementioned efforts, the incorporation of buffer-aided UAV with Q-learning for UAV's position optimization, specifically the investigation on OP minimization, remains insufficiently investigated. Motivated by the above, we proposed a Q-learning approach for optimizing UAV relay positioning, augmented by a buffer-aided UAV relaying scheme to minimize outage probability through managing the data flow while achieving optimal positioning through the Q-learning approach. As a result, the potential for jointly enhancing both large-scale by reducing (significant path loss) and small-scale multipath fading by minimizing (packet loss). The main

contributions of this paper are summarized as follows:

- 1) **Proposal of buffer-aided UAV relay:** We introduce a system model that integrates buffer capabilities with aerial propagation characteristics (UAV relay), which dynamically responds to diverse channel conditions and buffer configurations, offering an approach to managing the data flow, thereby enhancing UAV relay-aided communication reliability.
- 2) **RL optimization approach:** We introduce Reinforcement Learning (RL) approach based on Q-learning to intelligently optimize the UAV relay position with the aid of a buffer mechanism relative to the Base Station (BS) and Ground User (GU). to enhance overall communication reliability.
- 3) **Enhanced performance:** Our results demonstrate that the proposed buffer-aided UAV-Relay scheme based on Q-learning significantly benefits from joint optimization of both large-scale (significant path loss) and small-scale multipath fading (packet loss), leading to substantial OP reduction in UAV relay-aided communications, particularly when compared to the widely adopted buffer-aided fixed-UAV scheme and max-link scheme.

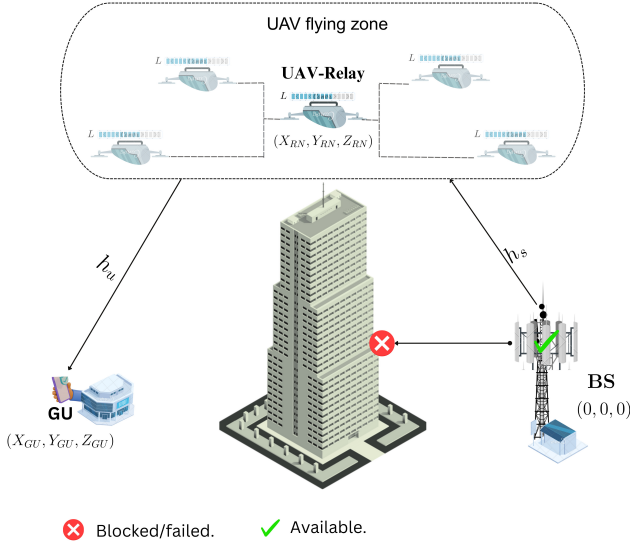


Fig. 1. System configuration of buffer-aided UAV relaying system, illustrating strategic node deployment.

II. SYSTEM MODEL

We hypothesize a scenario whereby the Base Station (BS) and Ground User (GU) are not directly connected, as portrayed in Figure. 1. Thus, the UAV serves as a Relay Node (RN) to ensure dependable communication between the BS and GU. Each node is equipped with a single antenna, and the RN operates in Half-Duplex (HD) strategy, utilizing the Decode-and-Forward (DF) approach. We presuppose that the time duration is divided into multiple intervals of equal duration, and the BS or RN transmits a packet with a constant power P in each slot. Assuming a 3-D Cartesian coordinate system, the locations of BS, RN, and GU are defined as $BS = (0, 0, 0)$, $RN = (x_{RN}[n], y_{RN}[n], z_{RN}[n])$, and GU

$= (x_{GU}[n], y_{GU}[n], z_{GU}[n])$, where (x_i, y_i, z_i) , $i \in (BS, RN, \text{ and } GU)$ denote the 3D Cartesian coordinates of the x , y , and z axes, accordingly, and the time slot $n \in \{1, \dots, N + 1\}$. The 3D distances between BS and RN, as well as RN and GU, can be calculated using the following formula:

$$d_S[n] = \sqrt{\xi_x[n] + \xi_y[n] + \xi_z[n]}, \quad (1a)$$

where, $\xi_x[n]$, $\xi_y[n]$, and $\xi_z[n]$ in Eq. (1a) indicate the differences of (x, y, z) coordinate between BS and RN and can be expressed as:

$$\xi_x[n] = (x_{RN}[n] - x_{BS}[n])^2, \quad (1b)$$

$$\xi_y[n] = (y_{RN}[n] - y_{BS}[n])^2, \quad (1c)$$

$$\xi_z[n] = (z_{RN}[n] - z_{BS}[n])^2. \quad (1d)$$

Moreover

$$d_U[n] = \sqrt{\Xi_x[n] + \Xi_y[n] + \Xi_z[n]}, \quad (2a)$$

here, $\Xi_x[n]$, $\Xi_y[n]$, and $\Xi_z[n]$ in Eq. (2a) demonstrate the variations of (x, y, z) coordinates between RN and GU which can be characterized as:

$$\Xi_x[n] = (x_{GU}[n] - x_{RN}[n])^2, \quad (2b)$$

$$\Xi_y[n] = (y_{GU}[n] - y_{RN}[n])^2, \quad (2c)$$

$$\Xi_z[n] = (z_{GU}[n] - z_{RN}[n])^2. \quad (2d)$$

The communication channels between the BS and the UAV (e.g., SR) and that associated with the UAV to GU (e.g., RU) are delineated utilizing a probabilistic Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) model. These probabilities of both LoS and NLoS can be articulated as [5]:

$$P_t^{\text{LoS}} = \frac{1}{1 + \mu e^{\nu(\theta_t - \mu)}}, \forall t \in \{SR, RU\}, \quad (3)$$

$$P_t^{\text{NLoS}} = 1 - P_t^{\text{LoS}}, \forall t \in \{SR, RU\}, \quad (4)$$

where μ and ν are constant parameters that are associated with a variety of environmental factors. The elevation angles between the BS and UAV can be calculated as:

$$\theta_s = \arctan \left(\frac{z_{RN} - z_{BS}}{\sqrt{(x_{RN} - x_{BS})^2 + (y_{RN} - y_{BS})^2}} \right), \quad (5)$$

as well as that between the UAV and GU can be calculated as:

$$\theta_u = \arctan \left(\frac{z_{GU} - z_{RN}}{\sqrt{(x_{GU} - x_{RN})^2 + (y_{GU} - y_{RN})^2}} \right). \quad (6)$$

In consequence of the probabilistic nature of LoS and NLoS, the small-scale fading among the BS-UAV and that between the UAV-GU can be characterized as [11]:

$$g_t = \sqrt{\frac{K}{K+1}} \sigma e^{j\theta} + \sqrt{\frac{1}{K+1}} \tilde{g}_t, \forall t \in \{SR, RU\}, \quad (7)$$

where K symbolizes the Rician factor, which is dependent on the elevation angle. As the elevation angle θ increases, the corresponding Rician factor and the LoS probability increase. And $\sigma e^{j\theta}$ denotes the deterministic value of LoS component, $\tilde{g}_i$ signifying the complex channel coefficient

of NLoS scattered component. Thus, the PDF of $|g_i|^2$ can be characterized as [5], [11]:

$$f_{\gamma(x)} = \frac{1}{2\sigma^2} \exp\left(-\frac{K[n]^2 + x}{2\sigma^2}\right) \cdot I_0\left(\frac{K[n]^2}{2\sigma^2} \sqrt{x}\right), x > 0, \quad (8)$$

where I_0 signifies the zeroth-order modified Bessel function of the first kind. The angle-dependent Rician factor was modelled as in [5], [11]:

$$K_t = k_{\min} e^{\left(\frac{2}{\pi} \ln\left(\frac{k_{\max}}{k_{\min}}\right) \theta_t \frac{\pi}{180}\right)}, t \in \{SR, RU\}, \quad (9)$$

wherein $k_{\max}$ and $k_{\min}$ symbolize the maximum and minimum Rician K-factor, respectively. The Signal-to-Noise Ratio (SNR) of the SR and that of RU links can be expressed as:

$$\gamma_t = \frac{P}{\sigma^2} |h_t|^2, \forall t \in \{SR, RU\}, \quad (10)$$

where P specifies the transmit power of both BS and RN, $|h_t|^2$ signifies channel gain from BS-to-RN and RN-to-GU. Concretely, the channel gain can be articulated as [5]:

$$|h_t|^2 = \frac{E}{d_t^{\Omega_t}}, \forall t \in \{SR, RU\}, \quad (11)$$

where the term E can be characterized as:

$$E = \frac{(4\pi f_c)^{-2}}{C}, \quad (12)$$

which reflects the constant channel attenuation and C is the speed of light. Furthermore, the factor that implies the aerial path loss exponent can be expressed as:

$$\Omega_t = \Omega^{NLoS} \cdot P_t^{NLoS} + \Omega^{LoS}, \forall t \in \{SR, RU\}, \quad (13)$$

alongside Ω^{LoS} and Ω^{NLoS} being influenced upon environmental characteristics. The attainable ergodic rates for both BS-UAV and UAV-GU links are delineated in [5] as follows:

$$R_{SR} = F \cdot \log_2(1 + \gamma_{SR} |g_{SR}|^2), \quad (14)$$

$$R_{RU} = F \cdot \log_2(1 + \gamma_{RU} |g_{RU}|^2), \quad (15)$$

where F signifies a constant factor which arises from the division of radio resources through the utilization of two time slots for the transmitting of each packet to the endpoint. $\gamma_t, t \in \{SR, RU\}$ is given in Eq. (10), and $|g_t|^2, t \in \{SR, RU\}$ delimited in Eq. (7) refers to an independent and identically distributed (i.i.d.) Random Variable (RV) following the non-central chi-square Probability Distribution Function (PDF) with two degrees of freedom. In other contexts $|g_t|$ is also referred as Rician fading and $\mathbb{E}[|g_t|^2] = 1$, where $\mathbb{E}[\cdot]$ signifies the expectation operator. In contrast, the outage probability of both links can be characterized as:

$$\begin{aligned} P_{SR} &= P_r(R_{SR} < R_{\min}) \\ &= 1 - Q_1\left(\sqrt{2K_{SR}}, \frac{\sqrt{2(K_{SR} + 1)\delta}}{\gamma_{SR}}\right), \end{aligned} \quad (16)$$

$$\begin{aligned} P_{RU} &= P_r(R_{RU} < R_{\min}) \\ &= 1 - Q_1\left(\sqrt{2K_{RU}}, \frac{\sqrt{2(K_{RU} + 1)\delta}}{\gamma_{RU}}\right), \end{aligned} \quad (17)$$

where $K_t, \forall t \in \{SR, RU\}$ denotes a Rician factor of the channel from BS to RN and that from RN to GU, respectively, $\delta = \frac{(2^{2R_{\min}} - 1)}{\Lambda d_t^{\alpha_t}}$, $R_{\min}$ denotes the target rate, $\gamma_t, \forall t \in \{SR, RU\}$ is an average SNR expressed in Eq. (10), and $Q_1(a, b)$ exhibits the first-order Marcum Q-function delineated as:

$$Q_1(a, b) = \int_a^b x \cdot \exp\left(-\frac{a^2 + x^2}{2}\right) \cdot I_0(ax) dx, \quad (18)$$

where $a = \sqrt{2K_t}$ and $b = \frac{\sqrt{2(K_t + 1)\delta}}{\gamma}$, $t \in \{SR, RU\}$. Considering buffer length L denotes the occupied buffer size in the n -th time slot, with initial status as 0 aided UAV relay to manage the data flow and minimize the OP among the links. To obtain better outage performance and enhance system resilience against signal degradation, the buffer relies on the quality of the links. Specifically, the OP can be expressed as the average of an outage indicator over n -th time slots, the indicator is determined by the following conditions:

$$I_{\text{out}} = \begin{cases} 1 & \text{if } (R_{SR} < R_{\min} \text{ and } R_{RU} < R_{\min}) \\ & \text{or } (R_{RU} < R_{\min} \text{ and } l = L) \\ & \text{or } (R_{SR} < R_{\min} \text{ and } l = 0), \\ 0 & \text{otherwise.} \end{cases} \quad (19)$$

the probabilities describe the OP based on the link availability and buffer status, which is the capability of accommodating various ranges of packet values, with a maximum size denoted as L . Thus, when no outage occurs (e.g., $I_{\text{out}} = 0$), the buffer level l is dynamically updated in non-outage cases based on the following:

- If $R_{SR} > R_{\min}$ and $R_{RU} < R_{\min}$ and $l < L$, then $l \leftarrow l + 1$.
- If $R_{SR} < R_{\min}$ and $R_{RU} > R_{\min}$ and $l > 0$, then $l \leftarrow l - 1$.
- If $R_{SR} > R_{\min}$ and $R_{RU} > R_{\min}$ and $l = 0$, then $l \leftarrow l + 1$.
- If $R_{SR} > R_{\min}$ and $R_{RU} > R_{\min}$ and $l = L$, then $l \leftarrow l - 1$.
- If $R_{SR} > R_{\min}$ and $R_{RU} > R_{\min}$ and $0 < l < L$, then:

$$l \leftarrow \begin{cases} l + 1 & \text{with probability } a, \\ l - 1 & \text{with probability } 1 - a, \end{cases} \quad (20)$$

where a denotes the threshold and can be obtained via one-dimensional search. Then the OP calculated as the expected value of the outage indicator over N time slots which can be characterized as:

$$P_{\text{out}} = \frac{1}{N} \sum_{n=1}^N I_{\text{out}}. \quad (21)$$

A. Problem formulation

In this section, the optimization problem is formulated as a problem of finding the optimal UAV's position (e.g., (x^*, y^*, z^*)) that can effectively minimize the OP. The problem can be characterized as:

$$(x^*, y^*, z^*) = \arg \min_{x, y, z} P_{\text{out}}(x, y, z).$$

It can be seen that the objective function $P_{\text{out}}(x, y, z)$ depends on both, R_{SR} and R_{RU} when the UAV is placed

at position (x, y, z) , which corresponds to the context in which their values are influenced by the position of the UAV due to path loss and fading. To solve this optimization problem, we adopt an RL exploiting a Q-table approach to iteratively determine the UAV's optimal position by exploring the discrete state space. A Q-table is a matrix that demonstrates the possible states and actions, enabling the agent to adopt the decisions according to its environment. The Q-table matrix is structured with dimensions of the possible states multiplied by the possible actions to allow the agent to select the best action for a given state. Thus, the agent can either select the action that yields the highest reward or can explore other actions to maximize the overall cumulative discounted reward. In this paper, the framework of the Q-table is unveiled as follows:

1) *State*: The state represented as $\mathcal{S}$ comprising all possible UAV's positions, where each state $s \in \mathcal{S}$ is a tuple (e.g., $s = (x, y, z)$) denoting the coordinates of the UAV in the x , y , and z directions. Each coordinate is discretized within a specified range (e.g., $x, y, z \in \text{grid points} = \{\xi_{\min}, \xi_{\min} + \Delta, \xi_{\min} + 2\Delta, \dots, \xi_{\max}\}$), where Δ symbolizes the grid step size. This creates a 3D grid with a finite number of states per dimension (e.g., $S = \left(\frac{\xi_{\max} - \xi_{\min}}{\Delta} + 1\right)^3$, $\forall \xi \in x, y, z$), where $(+1)$ is to ensure both boundary points ($\xi_{\min}, \xi_{\max}$). This enables the Q-learning to allocate each state to a row in the Q-table.

2) *Action*: The action symbolized as $\mathcal{A}$ represents the set of actions (e.g., $a \in \mathcal{A}$) where the UAV is capable of taking from any state (e.g., $s \in \mathcal{S}$). The agent, representing the UAV, can decide to maneuver the UAV in any direction (e.g., west, east, south, north, upward, downward, or remain stationary), with every motion leading to an aberration of $\pm\Delta$ in a single dimension as manifested in the following agent (e.g., UAV) movements matrix:

$$\mathcal{A} = \begin{bmatrix} \Delta x & 0 & 0 \\ -\Delta x & 0 & 0 \\ 0 & \Delta y & 0 \\ 0 & -\Delta y & 0 \\ 0 & 0 & \Delta z \\ 0 & 0 & -\Delta z \end{bmatrix} = 6, \quad (22)$$

these actions identify the subsequent condition reliant upon the present position.

3) *Reward Function*: The reward function $r(s, a)$ produces feedback depending on the action a performed in state s and it can be delimited as:

$$r = -\frac{1}{N} \sum_{n=1}^N P_{\text{out}}, \quad (23)$$

where P_{out} denotes the OP computed over N trials. Moreover, the reward is contingent upon the UAV's coordinates (x, y, z) and the selected method (proposed or Max-link), incorporating a negative value to mitigate OP, with penalties imposed for boundary violations.

B. Training Process

During the training process, the agent's learning begins with an initial state and undergoes many transitions through various states, finishing in a terminal state. Subsequent to performing each action, the Q-table undergoes an update.

Thus, the Q-learning approach is delineated by the subsequent update rule as [18]:

$$Q(s, a) \leftarrow (1 - \alpha)Q(s, a) + \alpha \left[r + \lambda_{\epsilon} \max_{a'} Q(s', a') \right], \quad (24)$$

where α denotes the learning rate, λ_{ϵ} symbolizes the discount factor, r represents the reward, and s' delineates the subsequent state following the execution of action a . The process involves N_{ep} number of episodes, each with a maximum of T_{max} steps, using an ϵ -greedy strategy with a minimum $\epsilon_{\min}$ and a decay factor λ_{ϵ} . After acquiring multiple Q-table entries through exploration, the agent embarks on exploitative actions in the following episodes. The exploration rate ϵ is updated to equilibrate exploratory and exploitative actions throughout episodes, this procedure is performed by updating ϵ in accordance with the following expression:

$$\epsilon_{t+1} = \max(\epsilon_{\min}, \epsilon_t \times \lambda_{\epsilon}), \quad (25)$$

where ϵ_t , ϵ_{t+1} , and $\epsilon_{\min}$, symbolize the current, updated, and minimum epsilon, respectively.

III. ALGORITHM COMPLEXITY ANALYSIS

The configuration of the Q-learning algorithm exploits the computational complexity of a specified 3D state space $\mathcal{S}$ within a specified range (e.g., $x, y, z \in \text{grid points} = \{\xi_{\min}, \xi_{\min} + \Delta, \xi_{\min} + 2\Delta, \dots, \xi_{\max}\}$), the operation of the Q-table, and OP calculation. $\mathcal{S}$ is designed to capture the spatial coordinates of the UAV, while the Q-learning approach optimizes its position. The environmental parameters SNR that influence the communication link quality and OP as well as L , which affect the OP and reward function, are fixed during the training process. Thus, the Q-table is characterized by dimensions $[\mathcal{S} \cdot \mathcal{A}]$, with $\mathcal{A} = 6$ actions and $\mathcal{S} = \left(\frac{x_{\max} - x_{\min}}{\Delta} + 1\right)^3 = (6 + 1)^3$ resulting in 7 grid points for each dimension in the 3D space, which leads to $(7)^3 = 343$ states and $O(\mathcal{S} \cdot \mathcal{A}) = (343 \cdot 6) \approx O(2058)$ entries. Both action selection with an ϵ -greedy strategy (e.g., $O(1)$) and Q-table updates (e.g., calculating the max of Q-value for the next s $O(\mathcal{A})$) are involved in each training step. The OP calculation utilizes Monte Carlo (MC) simulation with a number of iterations, each $O(1)$, resulting in $O(MC) = O(10^6)$ per step. For N_{ep} episodes and up to T_{max} steps per episode, the per-episode complexity is $O(T_{\text{max}} \cdot MC)$, and the overall training complexity is $O(N_{\text{ep}} \cdot T_{\text{max}} \cdot MC)$. Space complexity is dominated by the Q-table with a number of entries as $(\mathcal{S} \cdot \mathcal{A}) = (343 \cdot 6) = 2058$, each 8 bytes, requiring approximately 16.5 KB of memory.

IV. RESULTS AND DISCUSSION

In this section, we present the simulation results to evaluate the performance of the buffer-assisted UAV relay system. Unless specified otherwise, we summarized the system parameters in Table I. For benchmarking, we postulate both approaches with fixed UAV position configuration and the UAV-based Q-learning utilizing the max-link schemes.

In Fig. 2 the OPs of three different approaches are investigated relation to a variety of L buffer sizes. As illustrated in Fig. 2, the approach with the fixed UAV's position configuration (e.g., the baseline 1 scheme) demonstrates

TABLE I
SYSTEM PARAMETERS

Parameter	Value	Description
<i>Environment Setup</i>		
$BS(x, y, z)$	$(0, 5, 0)m$	Base station coordinates
$GU(x, y, z)$	$(497, 50, 0)m$	Ground user coordinates
F_C	$2GHz$	Carrier frequency
C	3.8×10^8 m/s	Propagation speed of light
MC	10^6	Monte Carlo simulation
R	$1bit/s/Hz$	Target rate
α_{LoS}	2.0	LoS Path-Loss Exponent
α_{NLoS}	1.5	NLoS Path-Loss Exponent
A, B	1.136, 11.95	Channel parameters
S, D	DF, HD	Strategy and duplexing technique
(K_{min}, k_{max})	5, 15	Rician factors
<i>Q-Learning Configuration</i>		
N_{ep}	100	Number of episodes
T_{max}	30	Maximum steps per episode
Δ	$25m$	Grid step size
N	10^6	Number of trials
α	0.2	Learning rate
ϵ	0.5	Initial exploration rate
ϵ_{min}	0.01	Minimum exploration rate
λ	0.99	Discount factor
$\mathcal{A}$	6	Action space size

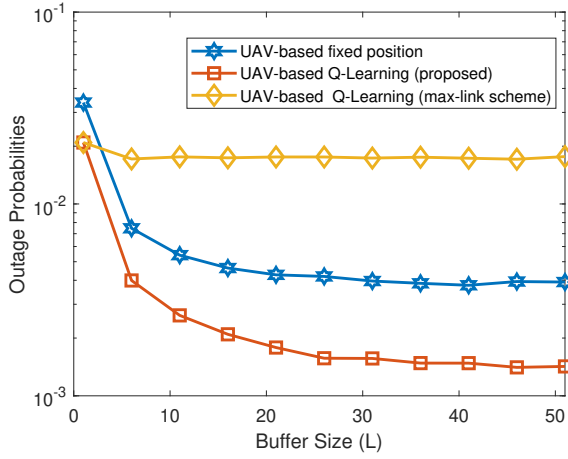


Fig. 2. Outage probabilities versus buffer sizes, comparing the proposed UAV-based Q-learning, Max-Link scheme, and fixed UAV's position baseline.

the highest initial OP of approximately $10^{-1.5}$, which then starts to decrease as the L buffer sizes increase by roughly $10^{-2.5}$. Conversely, the strategy of the UAV-based Q-learning approach (e.g., the proposed strategy) attains a considerably reduced OP along with various L sizes, commencing at approximately $10^{-1.8}$ and then diminishing to about $10^{-2.8}$, demonstrating enhanced dependability and superior UAV positioning optimization. Among the three approaches, the UAV-based Q-learning utilizing the max-link scheme (e.g., the baseline 2 scheme) maintains a fairly steady OP of approximately 10^{-1} across various L size ranges, achieving the worst effectively among the three schemes. These findings imply that the proposed Q-learning method efficiently decreases OP by dynamically optimizing the UAV's position in real-time, outperforming both baseline schemes.

The evaluation of the UAV-based positioning strategies is shown in Fig. 3, highlighting the comparison of the proposed UAV-based Q-learning with both baseline scheme. However, the proposed Q-learning method manifests a rapid convergence. Over the first 20 training rounds (e.g., episodes), the total rewards rising from nearly (e.g., -0.35)

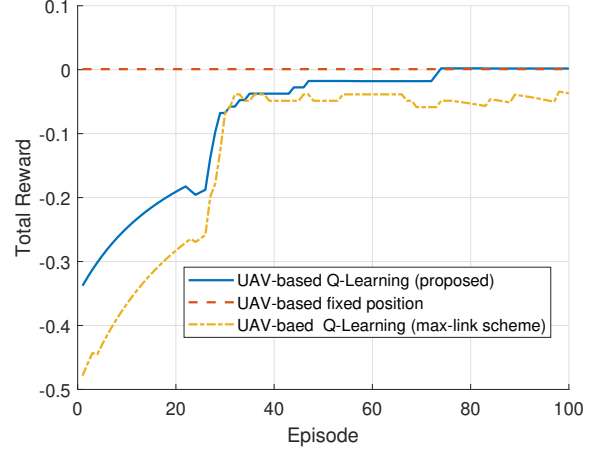


Fig. 3. Total reward versus episode, which reflects the proposed UAV-based Q-learning, Max-Link scheme, and fixed UAV's position baseline.

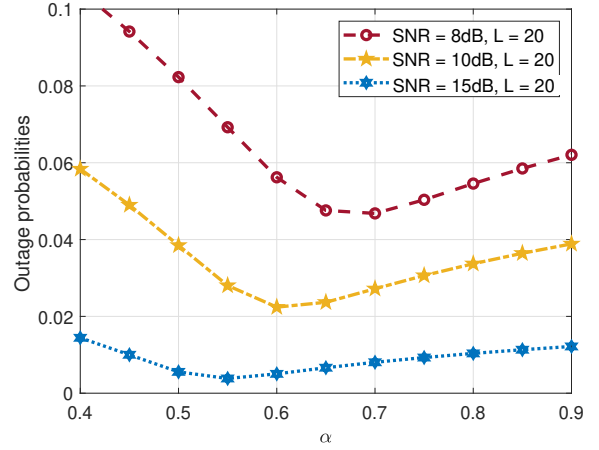


Fig. 4. Outage probabilities versus different thresholds α , manifesting the optimal α with various SNR values.

to almost 0, before stabilizing with minor fluctuations between 0 and -0.1 over 100 training episodes. This indicates the proposed strategy adapts the optimal UAV's position that minimizes the OP. By contrast, the fixed position approach maintained a constant reward of about 0 due to its static position. Rising from (e.g. -0.5) and elevating to approximately (e.g. -0.1) by episode 20, the UAV-based Q-learning approach that utilizes a max-link scheme implied slower enhancement compared to the proposed strategy. However, the scheme reduces the OP while remaining consistently lower than the proposed one. These findings highlight the higher efficiency of the proposed Q-learning strategy in optimizing the UAV's location for improving system reliability and consequently stressing its capacity to adapt and outperform both baseline approaches.

Fig. 4 illustrates the OPs of the system with respect to various α values and SNR, while L is set as $L = 20$. As manifest in Fig. 4, The outage probability is a clear dependency on the α value. Specifically, the optimal value of α when the SNR value is set to $8dB$ is observed to be approximately $\alpha = 0.7$, beyond which the OP rises gradually. When the SNR value is set at $10dB$, the optimal value of α is at 0.6. Whereas the optimal α value when

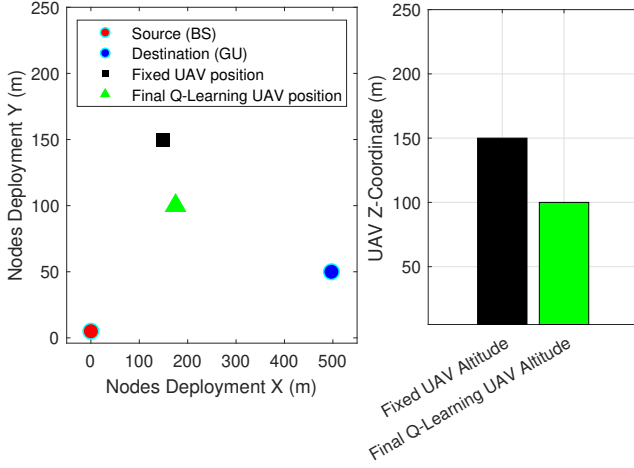


Fig. 5. 3-D Deployment of nodes, capturing the spatial configuration of the BS, GU, and UAV's Relay positions in fixed positions, and the Q-learning-based scheme.

the SNR value is set at $15dB$ is 0.55. Thus, selecting the optimal a value highlights its significance in influencing the overall system performance concerning OP.

Fig. 5 depicts a comparative assessment of the communication node deployment (e.g., BS, UAV-relay (RN), and GU) and positioning strategies in the x-y plane and UAV z-coordinate optimization using fixed positioning versus Q-learning. As illustrated in 5, the nodes deployment (x-y Plane) subplot, BS is located at (0, 5), UE at (497, 50), the fixed UAV's position at (150, 150), and the Q-learning optimized UAV position at approximately (100, 100). This spatial configuration demonstrates that the proposed Q-learning method adapts the UAV position approximately midpoint between the BS and UE, potentially optimizing the communication reliability. The UAV z-Coordinate comparison subplot additionally reveals that the fixed UAV is placed at a z-coordinate of 150 m, while the Q-learning-optimized UAV exhibits a lower elevation of 100 m, implying that the Q-learning approach refines the vertical position of the UAV to enhance link quality, thereby minimizing the OP. These visualizations demonstrate how dynamically adaptable the Q-learning approach is for both horizontal and vertical UAV positioning, therefore enhancing performance over the stationary fixed position approach.

V. CONCLUSION

In this paper, we explored the integration of a buffer mechanism-aided UAV relaying system and demonstrated the effectiveness of an RL-based Q-learning approach. Specifically, we integrated the buffer mechanism with the UAV relaying system to manage efficient data flow, thereby enhancing the system reliability and reducing OP. Furthermore, we contemplated a framework where UAV relay with the aid of a Q-learning approach conducted exploratory movements initially and exploitative motions in subsequent states to minimize the overall OP. Our simulation results manifested the proposed approach consistently outperforms the baseline schemes with fixed UAV's position and max-link, demonstrating the potential of exploiting Q-learning for UAV deployment in mobile wireless networks for

enhancing reliability, reducing OP, and convergence. Incorporating EE management strategies for the UAV could be an intriguing subject to explore further.

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